A09 ETHICAL AND SOCIETY IMPLICATIONS OF AI

CASE STUDY

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1. Introduction & Overview

For this case study, I'm defining Autonomous Vehicles (AVs) as systems that combine sensor fusion and machine learning to handle the road without us. To keep things organized, I'm sticking to the SAE's six levels of automation. Level 0 is basically what I grew up with, just a driver and a steering wheel, while Level 5 is the holy grail where a car drives anywhere, in any weather, with no steering wheel at all. Right now, most consumer tech like Tesla's Full Self-Driving is hovering at Level 2. The car handles the mechanics, but I'm still the one on the hook for oversight.

The industry is moving faster than I expected. As of March 2026, Waymo is already hitting 500,000 rides a week across ten cities. Then you have the Uber, Lucid, and Nuro collab from CES earlier this year, building a luxury robotaxi fleet with the Lucid Gravity. It's a clear sign we're moving past the science project phase and into actual infrastructure.

But I think we have to be realistic about the cost of this progress. While AVs can definitely cut down on accidents by removing human error, we can't ignore the ethical mess they create, things like decision-making algorithms, data privacy, and the reality of people losing their jobs. My goal for this assignment is to look past the hype and see how we actually manage those trade-offs.

The reality is that we are the problem on the road. Roughly 94% of accidents boil down to human errors such as fatigue, distraction, or just bad judgment like Tyger Woods! AVs are meant to strip away our bad habits and replace them with a more reliable AI solution.

I see the process broken into three systems. First, there's sensor fusion. It's like the car's senses working together: LiDAR maps the world in 3D, Radar tracks moving objects, and cameras handle visual recognition. You fuse that data so the car gets a much clearer picture than I ever could through a windshield.

Second is Deep Learning. I've been digging into how Convolutional Neural Networks (CNNs) handle object detection. The CNN takes those sensor feeds and decides if it's looking at a cyclist or a mailbox. It's impressive, but as a student, I'm always mindful that these models are only as good as their training data.

Finally, path planning pulls it all together. The AI takes everything it "sees" and makes instant calls on steering or braking. It's a constant loop: perceive, predict, and act. Doing that much math at 60 mph is the only way to actually get human failure out of the driver's seat.

2. Benefits & Positive Impact

The practical AV upside is difficult to dispute. For me, the biggest win is safety. In the United States alone, we lose more than 40,000 people a year to traffic accidents. To put that in perspective, it's as if a major plane crash occurred every other day. By removing the role of human error from the equation, we're not just improving commute times, we're pretty much solving a huge public health catastrophe that we've largely accepted as part of life.

And then there's efficiency, which is where the tech gets super interesting. I'm especially interested in platooning, where vehicles move in a closely aligned and synchronized line. Because the AI can react in a heartbeat, far faster than our own reflexes, these cars are able to remain close enough to greatly minimize air drag. The models say it could reduce fuel or energy consumption by as much as 10 percent. If that's a logistics company with a fleet of trucks, it's a big cut in overhead and carbon emissions.

Beyond its direct impact on the raw numbers, there's a human side to accessibility that I think gets lost amid all the technical upsides. For the elderly or people with visual and physical impairments, an AV isn't just a high-tech gadget, it's independence. It's the difference between being homebound and having a world of options open to you without needing to depend on a neighbor or family member for transportation.

Finally, we need to consider the environmental cost. If I'm honest, AI is just more well-behaved at the pedal than me. It has the ability to optimize acceleration and regenerative braking to get every ounce of range out of an EV battery. Pair that with less traffic congestion, as AI doesn't "rubberneck", we should start to see the overall footprint of our transportation system shrinking. It's a classic example of applying better logic to address a physical resource problem.

3. Ethical Issues

Part of this case study that I find most difficult to get my head around is the ethics behind AI in Autonomous Vehicles. It's one thing to optimize a CNN to detect an object better, it's another dilemma entirely to make that code decide what price to put on a human life when a crash is literally inevitable.

We talked about the Trolley Problem in an ethics class I took as a thought experiment, but for a developer it must be critical logic requirement. If a car has to choose between plowing into a concrete barrier, likely killing me, and the passenger, or a group of kids walking along on the sidewalk, what does the script actually run? I've been exploring the PHM-Ethics

2026 system, which employs a risk-weighting formula rather than just yes or no choices. Which is a solid step in the right direction but I'm still a little curious. You can't simply code your way to avoiding a moral catastrophe, and I'm not sure we should be imagining that we can.

And then there's the bias problem, I found a study from King's College London which showed detection models have a 7–10% higher miss rate for pedestrians with darker skin and young children. If your training data consists mostly of light-skinned adults in the sun, then when the model's confidence plummets in low light, it isn't an embarrassing glitch, it's a systemic failure. Safety would matter little if it is not shared equitably. It's a reminder of how objective these systems can really be.

Lastly, there's the privacy concern. These vehicles are basically mobile surveillance units. Between the LiDAR mapping and 360-degree cameras they're capturing massive amounts of data with each passing second. We're finally asking who actually owns that footage, as new 2026 privacy laws come in places like California and Connecticut. Is it mine, the manufacturer, or is it for the city? I've seen this report where you have companies like Waymo that share trip data with regulators and then others who treat it as if it's a trade secret. It makes me think we're really just swapping our personal privacy for a little smoother ride.

4. Societal Implications

I've been really reflecting on the human cost of this technology, particularly since I live in Houston and drive by the traffic on I-45 and 610 every day. We're not just talking about cool software, we're talking about the livelihoods of about two million long-haul truck drivers and a massive fleet of rideshare workers. Here in Houston, you have companies like Aurora Innovation already transporting their autonomous freight routes from Dallas to Houston. It's technically impressive to watch a Class 8 truck navigate that corridor, but from a practical standpoint, I'm concerned about what happens to the people behind the wheel. We can't just consider "disruption" like it's a line of code, we need a practical plan for how these workers transition.

On the other hand, the way our cities are designed could be completely transformed, and that would be something I'd like to see. Essentially all of Houston is just a huge expensive parking lot at this point. If we shift to shared AVs, we won't need all that concrete in the center anymore. That's even more prime land that could be turned into green spaces or actual affordable housing.

But the real obstacle for society isn't tech, it's the trust gap. I was reading a report which discovered that only 13% of drivers currently trust riding in a fully self-driving car. That's a huge problem. People feel like they are losing control, and high-profile accidents contribute to that narrative. As a student in AI, I believe we have to get this down pat, if we fail to demonstrate that the system can be consistently safer than that of a human, then that trust is not coming back.

5. Risks & Challenges

I know that even the best code reaches a brick wall when facing real-world problems. We can mimic millions of miles, but the real world is too messy and unpredictable. The biggest obstacle is technical edge cases. Consider the massive San Francisco blackout back in December of 2025. Waymo's fleet essentially turned into expensive paperweights because the vehicles were unable to deal with dark traffic signals. Then there's the situation in Austin this year, with Waymo cars still failing to see stopped school buses. If a system can't recognize a massive yellow bus with flashing lights, I'm not ready to trust it in a Houston flash flood or on an unpaved country road in Fort Bend County.

We are faced with the liability Black Box. If a car crashes, who is to blame, the developer? The sensor? The dataset? Texas is already moving in the direction of a 2026 product liability model, with the heat on the developer and manufacturer, not the passenger.

6. Recommendations

To wrap up this case study, I have some actionable recommendations. It's easy to get lost in the cool side of the tech, but as an AI student I have come to realize that the code is only half the battle and what happens here is about how we build an ecosystem around it.

To start, we need to overcome this patchwork of state by state rules. Currently, a car's legal behavior can practically change just by crossing a state line. We need a consistent federal path under NHTSA that establishes clear safety standards. It's the only way to guarantee that a robotaxi in Houston exercises the same safety logic as one in Phoenix.

I believe that we need to require Explainable AI in AVs. We need audit trails that enable investigators to trace a particular decision back to the sensor input that caused it. Particularly with the EU AI Act coming into force this year, it's clear that if you can't explain why a model made a mistake, you shouldn't be deploying it on public roads.

And finally, how these systems are going to interact with the real world during an emergency needs to be standardized. It's good to see the Texas DMV requiring a First Responder Interaction Plan, like how do you instruct a driverless vehicle to pull over or shut down if there's a crash? It's also time to shed some light on bias. I've been examining the IEEE 7003-2025 standards for algorithmic bias, we should be enforcing this type of bias audits or something similar on every training dataset before a car ever gets to the pavement.

7. Conclusion

In short, it's obvious that Autonomous Vehicles are no longer an interesting engineering solution but more of a new paradigm shift. I'm genuinely impressed to see the technical development of the Level 4 level that we've reached by early 2026, especially as Waymo hits its stride and the Uber-Lucid partnership brings more of a commercial aspect to what's possible. The potential to save 40,000 people a year and finally take our cities back from unending parking lots is huge.

But as a student who views this code, I can't ignore the "black box" problems. If you ask me, while we can't control all the edge cases, or perhaps the speed of how things evolve in the industry itself, we definitely have control over the standards that we are setting. We need federal regulation, explainable AI for transparency, and a commitment to bias audits before we expand this field further. We could get all of the sensor fusion correct in the world, but if we don't get the ethics right behind it, there will never be enough to close that trust gap.

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